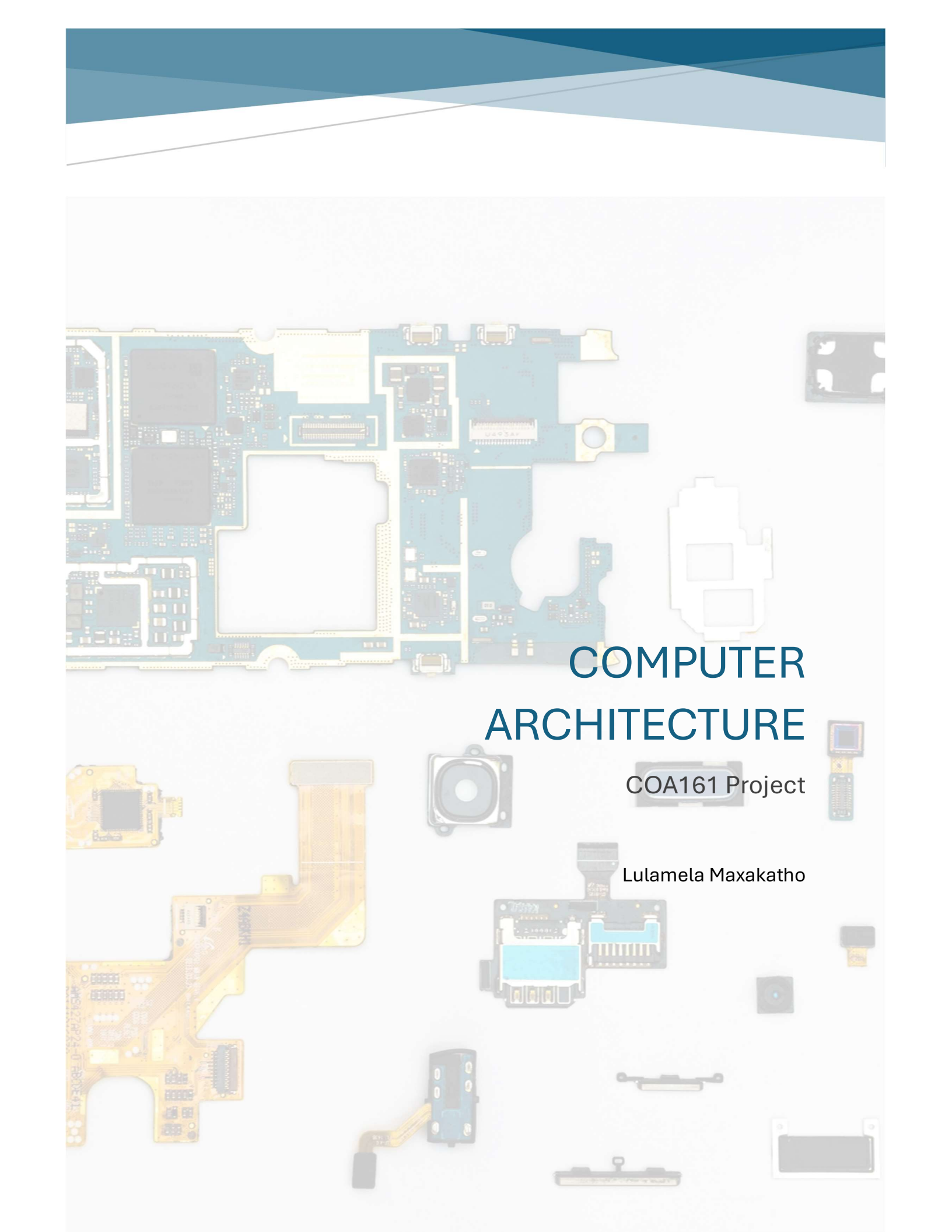


The background of the slide is a collage of various computer hardware components, including a large blue printed circuit board (PCB) with multiple chips, a white plastic component, a yellow flexible circuit board, a camera module, a small display screen, a USB connector, and other smaller electronic parts. The components are arranged in a scattered, artistic manner.

COMPUTER ARCHITECTURE

COA161 Project

Lulamela Maxakatho

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1. PC Build for an Online Gamer:

Here is the link to the PCPartPicker online game PC build: <https://pcpartpicker.com/list/m6Cx4M>

I designed a PC that will be suitable for an online gamer. I kept the budget for this PC build at a more realistic level. Not too expensive, but not too cost saving neither.

Hardware recommendations -

a. Processor (CPU):

For a gamer who enjoys the challenges of online gaming you'll want a processor that delivers high single-threaded performance and fast clock speeds, as these features are most critical for gaming performance.

Keeping an affordable budget in mind, I have chosen the Intel Core i5-13600K. It's a powerful mid-range CPU, excellent for gamers seeking high performance and flexibility.

Some key features include:

- **Graphics:** Comes with basic integrated graphics, although a GPU is recommended for the best possible performance
- **Intel Thread Director:** It optimizes the workload distribution across the P-cores and the E-cores, ensuring constant, uninterrupted performance.
- **Base Clock Speed:** 3.5GHz allows for higher speed under greater load

b. Memory (RAM):

For the memory of this PC, I have gone with the Corsair Vengeance DDR5-5200. It is a great option for those seeking reliable performance at a less bank-breaking price. It gives significant speed and bandwidth improvements over the DDR4 models, resulting in faster load times, overall system responsiveness, and a compact design as an added bonus.

Some key features include:

- **Capacity:** 16 GB (2 x 8 GB)
- **Speed:** 5200 MHz
- **Compatibility:** Optimized for Intel and AMD platforms

c. Hard Disk:

The Crucial P3 Plus 2 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive is a great choice as it grants fast read/write speeds, which means quicker boot times and faster game loading, yet it's still quite affordable compared to the other models. Its compact design offers easy install onto the motherboard.

Some key features include:

- **Capacity:** 2 TB
- **Interface:** PCIe 4.0 NVMe
- **Read Speed:** Up to 5000 MB/s
- **Write Speed:** Up to 4200 MB/s

d. Display:

The Dell G2724D Monitor has a 27-inch screen accompanied by 1440p resolution, which is great for extra clarity and performance. The 165Hz refresh rate ensures smooth gameplay, especially beneficial for fast-paced games. Although it's a bit pricey, it has amazing reviews on Amazon and offers reliable constant screen time.

Some key features:

- **Screen Size:** 27"
- **Resolution:** 2560 x 1440
- **Refresh Rate:** 165 Hz

e. Mother Board:

Naturally I chose a motherboard compatible with the Intel Core i5-13600K processor. The Asus ROG (Republic of Gamers) STRIX Z790 is an ATX motherboard designed for Intel's 12th and 13th generation processors. It features amazing audio quality, comprehensive cooling options, and has an eye-catching aesthetic.

Some key features include:

- **Memory Support:** It supports DDR5 memory, up to 5333 MHz (OC), allowing for high-speed performance in gaming.
- **Connectivity:** Includes faster wireless connectivity, alongside multiple USB 3.2 ports, and Thunderbolt 4 support.
- **Expansion & Storage:** The board features PCIe 5.0 slots for next-gen GPUs and M.2 slots with PCIe 4.0 for fast NVMe SSDs.

What is Thunderbolt 4?

"Thunderbolt™ 4 technology is a connectivity standard that delivers a constant 40 Gbps bandwidth for data, power, and video signal transfers between devices. Thunderbolt™ connections ensure an optimal experience by dynamically allocating total bandwidth between video and data." (Intel, n.d.)

f. Power Supply:

The Corsair RM750e (2023) 750 W 80+ Gold Certified Fully Modular ATX Power Supply is a high-quality power, reduced energy and heat consuming, quiet power supply unit designed for gamers.

Some key features include:

- **80+ Gold Certification:** Ensures high energy efficiency (up to 90% efficiency), reducing energy consumption and heat, leading to lower electricity bills and quieter operation
- **Fully Modular Design:** Improved cable management and airflow inside your PC case
- **Protection Measures:** Comes with a range of protection measures, including over-voltage, under-voltage, and short-circuit protection

g. Operating System and Software Required:

Windows 11 is the last operating system that offers a range of features for gaming needs:

Some key features include:

- **Game Mode:** Prioritizes system resources for gaming, reducing background activity to ensure the best performance during gameplay.
- **Direct Storage:** A feature that reduces load times by allowing games to directly access the storage on NVMe SSD's.
- **User Interface:** It provides a smooth, simple navigation system to quickly access games when logging onto your computer.

h. Input and Output Devices:

Keyboard:

The Redragon K552 Wired Gaming Keyboard. An affordable yet durable option equipped with mechanical switches that provide tactical feedback and a satisfying click. As well as some "always needed" customizable RGB lighting.

Mouse:

The Logitech G305 LIGHTSPEED Wireless/Wired Optical Mouse. You have the option to either struggle with a wire or enjoy a wireless experience. Paired with its light weight and high DPI (Dots Per Inch) sensor, it's just the perfect fit for any gamer.

Headphones:

The Razer BlackShark V2 X 7.1 Channel Headset. It's not the most amazing headset out there, but it's comfortable, gives clear communication, and immersive audio at an affordable price.

Speakers:

The Creative Labs Pebble 2.0 4.4 W Speakers. It's a decent, affordable set of speakers with a sleek, minimalistic compact design. It delivers 4.4W of RMS (Root Mean Square) power with an elevation of 45 degrees for better sound quality.

Webcam:

The Logitech BRIO Ultra HD Pro Webcam. If you ever want to stream content and have your face in the video, this webcam is a great option. It offers clear 4K Ultra HD resolution and adjusts the image based on the lighting conditions for the best high-quality output.

Steps to install and prepare the hard disks:

Preparation steps -

Part 1: Physically Installing the SSD

Locate the M.2 Slot:

- Identify the M.2 slot on your Asus ROG STRIX Z790 motherboard. The motherboard typically has several M.2 slots, often found near the PCIe slots.

Insert the M.2 SSD:

- Gently insert the SSD into the M.2 slot at an angle of about 30-45 degrees. Ensure the notch on the SSD aligns with the key in the slot.
- Push the SSD down until it's parallel with the motherboard.

Secure the SSD:

- Use the screw provided with your motherboard to secure the SSD in place.

Part 2: Preparing the Bootable Flash Drive

Create a Bootable Windows 11 USB Drive:

- On another computer, download the Windows 11 installation media creation tool from the [Microsoft website](#).
- Run the tool and follow the instructions to create a bootable USB drive.

Part 3: Installing Windows on the New SSD

Enter BIOS Setup:

- Once you have completed assembled the PC, press the power button.
- Immediately press the Delete key (or **F2** key) repeatedly during startup to enter the BIOS. As mentioned, "Timing is crucial".

Set the Boot Order:

- In the BIOS, go to the Boot tab.

- Set your USB drive as the first boot device.

Save and Exit BIOS:

- Save all the changes made and then exit the BIOS.
- Your PC will then restart and boot from the USB drive. This might take a few minutes.

Windows Installation:

- Follow the instructions provided through the prompts
- Choose Custom: Install Windows only (advanced).

Partition the SSD:

- You'll see a list of drives. Locate your Crucial P3 Plus SSD.
- If the drive is unallocated, click on it and choose New to create a new partition. Windows will automatically allocate the necessary space.
- Select the newly created partition and click Next.

Install Windows:

- Windows will begin installing on the SSD. This process will take some time.
- Your PC will restart a few times during the installation process.

Complete the Setup:

- Follow the on-screen instructions to complete the Windows setup, including signing in with your Microsoft account, setting your region, and configuring other settings.

2. PC Build for a 3D Designer:

Here is the link to the PCPartPicker 3D Designer PC build: <https://pcpartpicker.com/list/FmbJ6D>

Hardware recommendations -

a. Processor

Recommended: AMD Ryzen 9 7950X

Justification: For 3D designers, the AMD Ryzen 9 7950X is a strong CPU that deliver great performance in programs like Blender, AutoCAD, and SolidWorks.

b. Memory

Recommended: 64GB DDR5 5200MHz (G.Skill Trident Z5)

Justification: Large models and intricate textures can be handled by the memory-intensive 3D modelling software with 64GB of DDR5 RAM, avoiding slowdowns.

c. Storage

Recommended: Primary SSD: 2TB NVMe SSD (Samsung 990 Pro)

Secondary HDD: 4TB 7200 RPM HDD (Western Digital Black)

Justification: Faster load times and more seamless performance are achieved by the NVMe SSD's quick read/write speeds for the operating system, applications, and ongoing projects. Enough space is provided by the secondary HDD for major projects, backups, and frequently visited files.

d. Display

Recommended: 27-inch 4K UHD Monitor with 100% sRGB and Adobe RGB coverage (BenQ PD27505U)

Justification: For a 3D designer to see their work in high detail and guarantee proper colour representation—which is crucial in design work—they need a 4K UHD display with accurate colour reproduction.

e. Motherboard

Recommended: ASUS ROG Crosshair X670E Hero

Justification: High-performance computing requires excellent VRM cooling and power delivery, and this motherboard is designed to support the newest processors and PCIe 5.0. Additionally, it has enough USB ports for peripherals and many M.2 slots for SSDs.

f. Power Supply

Recommended: 850W 80+ Gold Certified PSU (Corsair RM850x)

Justification: An 80+ Gold certified PSU with a high wattage ensures longevity, efficiency, and stable power delivery—all essential for a high-performance build, particularly one that includes a potent CPU and possibly a high-end GPU.

g. Operating System and Software

Recommended: Windows 11 Professional

Software: Adobe Creative Cloud, Autodesk 3ds Max, Blender, and Microsoft Office Suite

Justification: Professional design software can be operated on a reliable and flexible platform with Windows 11 Professional. Essential tools for 3D modelling, rendering, and content development are included in the list of software.

h. Input and Output Devices

Recommended:

Keyboard: Mechanical Keyboard (Logitech G915)

Mouse: High-Precision Mouse (Logitech MX Master 3)

Justification: A mechanical keyboard offers a smooth and responsive typing experience, while a high-precision mouse guarantees precise control—an essential feature for 3D design work.

Steps to install and prepare the hard disks:

Preparation steps -

1. Create a Bootable Hard Drive:

- Purchase the Windows 11 ISO directly from Microsoft.com.
- To make a bootable USB drive, use an application such as Rufus. Make sure there is at least 8GB of space on the HDD.

2. Install the Operating System:

- Choose the bootable USB drive from the BIOS boot menu to start the computer from it after inserting it.
- On the primary NVMe SSD, install Windows 11 by following the on-screen instructions.

3. Partition and Format Secondary HDD:

- To access Disc Management once Windows 11 has been installed and launched, right-click on the Start menu and choose "Disc Management."
- Find the secondary HDD, then use the right-click menu to choose "Initialise Disc."
- Use the "New Simple Volume" option when you right-click the unallocated space to partition and format the drive as NTFS.
Give it a label and drive letter for simple identification.

4. Install Drivers and Software:

- Install the most recent drivers for all hardware components from the websites of the corresponding manufacturers after installing the operating system.
- Install any software that the designer requires, including Blender, Adobe Creative Cloud, and other programs.

5. Backup and Recovery Setup:

- To guarantee that the designer's work is consistently backed up to the secondary HDD or an external storage solution, set up a backup solution using Windows Backup or an independent tool. creator.

6. Optimization and Final Checks

- Make sure the BIOS has the SSD designated as the primary boot drive.
- Optimise the speed of Windows by modifying power settings and turning off unused starting apps.
- Conduct several stress tests and benchmarks to confirm the system's performance and stability.

3. PC Build for a Primary School Secretary:

Here is the link to the PCPartPicker primary school secretary PC build:

<https://pcpartpicker.com/list/bqNbBL>

The following design is for a PC that will be used by a secretary at a primary school. The PC has been designed bearing in mind the responsibilities of a secretary as well as the day-to-day work that they do. The PC's design prioritizes affordability and practicality.

Hardware recommendations -

a. Processor (CPU):

Much of the work of a secretary involves a lot of multitasking, therefore it is important that the PC offers reliable performance for the everyday office tasks that are within the secretary's occupation. This includes keeping track of student's records and files, schedule and confirm appointments, keep track of events as well as compiling and submitting bills. Word processing, browsing on the web and accessing emails are some of the critical features the PC should be able to do repeatedly and seamlessly.

The Intel Core i5-11400F is the chosen CPU for this PC. The Intel Core i5 offers strong performance, making it the best pick for a secretary who's work involves multitasking where applications like word and spreadsheets are greatly used. Some key features making this the ideal CPU are:

- **Includes CPU cooler:** it provides adequate cooling for typical office tasks and reduced noise. Perfect for the office atmosphere.
- **Simultaneous Multithreading (SMT) (hyper-threading):** Allows for multiple threads to be executed simultaneously, ideal for multitasking.

b. Memory (RAM):

For typical office tasks, the Corsair Vengeance LPX 8GB DDR4-2400 is both cost-effective and sufficient for tasks like browsing, word, and running multiple email applications without issues in performance. Its 5-star rating is justification to this. The DDR4 RAM is an improved version in terms of faster data transfer and power efficiency compared to the older DDR3 memory. Key features:

- **Heat Spreader:** Helps dissipate heat. Ensuring good performance even over extended use. Ideal for the secretary as they work with the PC almost all day.
- **Single Stick Configuration (1 x 8 GB):** In case the secretary's workload increases overtime it leaves additional space for other RAM slots.
- **Compatibility:** Compatible with a variety of motherboards.

c. Hard Disk:

Keeping in mind the multiple tasks that a secretary works on, including multiple records of students, a huge influx of emails received and compiling bills, the Seagate IronWolf Pro 24 TB 3.5" 7200 RPM is more than suitable with its massive storage of 24 TB. Its massive storage capacity ensures more than enough space for storing all necessary files as well as back up each file with no worry of running out of space. Key features:

- **Speed of 7200 RPM:** Faster read/write speed resulting in quicker access to files quick overall responsiveness.

d. Display:

Secretaries work with multiple applications simultaneously. A good monitor like the Asus VA24EHF with a 23.8-inch screen and 1920 x 1080(Full HD) is ideal as it provides ample space for office tasks like viewing documents, emails, and spreadsheets without taking majority of the desk space. The Full HD resolution makes multitasking much easier, allowing viewing for multiple windows side by side. The Full HD resolution also offers clear pictures and sharp texts. Some key features:

- **Refresh Rate:** The 100Hz refresh rate is responsible for the smoother visuals and transitions. It is good for eye care as it reduces eye strain caused by prolonged use.
- **Panel Type: IPS:** good color accuracy and wide viewing angles. Ideal for secretaries to view from different angles.

e. Motherboard:

The motherboard compatible with the Intel Core i5 is the Asus PRIME B560M-A/CSM Micro ATX LGA1200 Motherboard. The motherboard has a Micro ATX form factor which is compact, making it suitable for a mid-range office PC. This compactness also means that it does leave some space in the PC case. Offers 4 memory slots for additional storage. This motherboard also has integrated cooling fan headers, making it ideal especially for a secretary's work where the PC might be running for long hours. Some key features:

- **Memory Support:** It Supports DDR4 RAM, offers flexibility even for future updates and it is ideal for supporting multiple tasks efficiently.

f. Power Supply:

The Seasonic FOCUS Gold 550 w 80+ Gold Certified Semi-modular ATX Power Supply is the suitable power supply for this PC as it is reliable and energy efficient. Its compact size also makes it ideal as it can fit in smaller cases.

- **550W Power Output:** Ideal for most office PCs with multiple peripherals. It provides enough power to run the system.
- **80+ Gold Certification:** Highly efficient. Ideal for office spaces as it generates less heat, which contributes to a cooler quieter environment.

g. Operating System and Software Required:

Microsoft Windows 11 is suitable for office use. It offers wide compatibility with office software and peripherals. It is compatible with the Intel core i5 and the Asus PRIME B560M-A/CSM Micro ATX LGA1200 Motherboard.

Justification: Handles office tasks such as spreadsheets, word processing, email management and entering data.

h. Input and Output Devices:

Keyboard: Logitech K120 Wired Standard Keyboard.

Justification: Comfortable and quiet typing. It has a thin profile making it ideal for desk space. Sturdy and adjustable tilt legs. It has a plug-and-play USB connection.

Mouse: Logitech G305 Lightspeed Wireless/Wired Optical Mouse.

Justification: It is both a wired and wireless mouse.

Steps to install and prepare the hard disks:

Preparation steps -

1. Prepare Installation Media

- **Make a USB drive that can be booted:**
 - Create a bootable USB drive with Windows 11 using Rufus or the Windows Media Creation Tool.

2. Install Hardware

- **CPU:** Insert the Intel Core i5 into the motherboard's CPU socket.
- **RAM:** Place the RAM modules in the proper slots.
- **Storage:** Insert the NVMe SSD into the motherboard's M.2 slot.

- **Power Supply:** Mount the power supply to the motherboard and the storage and mount it in the case.

3. Install Windows 11

- **Boot from USB:**
 - Restart the computer after inserting the bootable USB drive. Set the USB drive as the first boot device and enter BIOS/UEFI using the DEL or F2 keys.
- **Install OS:**
 - For Windows 11, follow the on-screen instructions. If necessary, format the NVMe SSD before using it as the installation drive.
- **Complete Setup:**
 - Follow the on-screen instructions to complete the Windows setup, which includes creating a user account and setting up regional settings.

4. Post-Installation

- **Install Drivers:**
 - Visit the website of the motherboard manufacturer to obtain and install drivers.
- **Update Windows:**
 - Check the operating system's state by running Windows Update.
- **Install Software:**
 - Set up the software you need, like Microsoft Office 365.